INCH-POUND
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DETAIL SPECIFICATION

CLOTH, WIND RESISTANT OXFORD, NYLON AND COTTON, WATER REPELLENT TREATED

This document is approved for use by all Department and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one (1) type of cloth, wind resistant oxford, nylon and cotton, with a durable water repellent treated.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government Documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil

AMSC N/A FSC-8305

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Product Identification Act
Title 40 CFR - Protection of Environment

(Copies of this document are available online at https://www/ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter Method
AATCC TM15 - Test Method for Colorfastness to Perspiration
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM20A - Test Method for Fiber Analysis: Quantitative
AATCC TM22 - Test Method for Water Repellency: Spray Test
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM70 - Test Method for Water Repellency: Tumble Jar Dynamic Absorption
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM118 - Test Method for Oil Repellency: Hydrocarbon Resistance Test
AATCC TM127 - Test Method for Water Resistance: Hydrostatic Pressure
AATCC TM135 - Test Method for Appearance of Apparel and Other Textile End Products after Home Laundering

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(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737	-	Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424	-	Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D1683/D1683M	-	Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D3775	-	Standard Test Method for Warp (End) and Filling (Pick) Fabric Count of Woven Fabric
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.1.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Fibers.

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3.4.1.1 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a normal cut length of 1-1/2 inches and a round cross section with a single denier of 2.25 - 2.5 (0.25 - 0.28 Tex). The contractor shall submit the fiber producer's certification that each lot of nylon staple used conforms to the requirements specified herein.

3.4.1.2 Cotton. The cotton shall be carded.

3.4.2 Weave. The cloth shall be an oxford weave.

3.4.3 Yarn. The yarns shall be a blend of nylon and cotton, drawn and spun into single yarns for both warp and filling. The finished cloth shall contain 50 (± 5) percent nylon and the remaining percentage cotton, based on the dry weight of the desized specimen, when tested as specified in 4.5.

3.5 Color. The color shall be as specified in the contract or procurement documents (see 6.2).

3.5.1 Visual shade matching. The color and appearance of the dyed cloth shall match the standard sample when tested as specified in 4.5.

3.5.2 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be "Free" of labile sulfur when tested as specified in 4.5.

3.5.3 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements.

Colors Evaluation	Laundering Color Change and Staining grade, (3 cycles),(min.)	Light (after 40 AFU or 170 kJ/(m^2nm)@420 nm), grade, (min.) 1/	Perspiration (acid & alkaline) Color Change and Staining grade, (min.)	Crocking (Dry/Wet) grade, (min.)
All colors	3.5	3.5	3.5	3.5

1/ AFU: AATCC Fading Units.

3.6 Physical requirements. The finished cloth shall conform to the requirements in Table II when tested as specified in 4.5.

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TABLE II. Physical requirements.

Characteristic	Requirement
Weight, oz./sq.yd. Min. Max.	 6.3 7.1
Yarns per inch, (min.) Warp Filling	 130 48
Breaking strength, pounds, (min.) Warp Filling	 240 90
Tearing strength, pounds, (min.) Filling	 3.5
Dimensional change, percent, (max.) 1/ Warp Filling	 3.5 2.5
Air permeability, cu.ft/min./sq.ft., (max.) 2/	4.5
Seam efficiency, percent, (min.)	80
pH	5.5 – 8.5

1/ The dimensional change requirement refers to both shrinkage (-) and growth (+).
2/ After 5 (five) launderings.

3.7 Finish. The cloth shall be singed, desized, scoured, mercerized, and dyed. The cloth shall be given a durable water repellent treatment.

3.7.1 Water repellency. The cloth shall be given a durable water repellent treatment. The cloth shall meet the requirements in Table III when tested as specified in 4.5.

TABLE III. Water repellency.

	Hydrostatic pressure, cm		Dynamic absorption (percent)		Spray rating 3/
	Min. lot avg.	Min. 1/	Max. lot avg.	Max.2/	
Initial	30	25	20	25	90,90,80
After 5 launderings	30	25	20	25	----

1/ No individual specimen shall exceed the specified minimum.
2/ No individual specimen shall exceed the specified maximum.
3/ The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than the specified ratings when tested as specified in 4.5.

3.7.2 Resistance to organic liquid. The finished cloth shall show no wetting by n-tetradecane initially and after 5 launderings when tested as specified in 4.5.

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3.8 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7.

3. 9 Face identification. The warp side shall be identified as the face side by applying stamping on that side of the cloth with the word "FACE" at each end of the roll.

3.10 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of woven selvages when fly-shuttle looms are used, and exclusive of the selvage when shuttles looms are used.

3.11 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. The cloth shall be put up on full rolls as specified in 5.1.

3.12 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Conformance inspection (see 4.3.2).

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified, or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected for appearance, color, the defects of 4.4 and tested for the characteristics as specified in 4.5 through 4.7.

4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or procurement documents, except where otherwise indicated (see 6.2).

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4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Type I of MIL-STD-3064.

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yard full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Material testing.

Characteristic	Requirement Reference	Test method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.4)
Weave	See 3.4.2	Visual
Fiber content	See 3.4.3	AATCC TM20A
Visual shade matching	See 3.5.1	See 4.6.1
Labile sulfur	See 3.5.2	See 4.6.2
Colorfastness: Laundering Light Perspiration Crocking	See Table I	 AATCC TM61 Test 3A 1/, 2/ AATCC TM16.3, Option 3 3/ AATCC TM15 2/ AATCC TM8 4/

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TABLE IV. Material testing – Continued.

Characteristic	Requirement Reference	Test method
Weight	See Table II	ASTM D3776/D3776M
Yarns per inch	See Table II	ASTM D3775
Breaking strength	See Table II	ASTM D5034
Tearing strength (filling)	See Table II	ASTM D1424
Dimensional change	See Table II	AATCC TM135, (3),(III),(Aiii)
Air permeability: After five (5) launderings	See Table II	AATCC TM135, (3),(V),(Aiii) & ASTM D737
Seam efficiency	See Table II	ASTM D1683/D1683M
pH	See Table II	AATCC TM81
Water repellency:		
Hydrostatic pressure:	See Table III	
Initial		AATCC TM127
After 5 launderings		AATCC TM135, (3),(V),(Aiii) & AATCC TM127
Dynamic absorption:	See Table III	
Initial		AATCC TM70
After 5 launderings		AATCC TM135, (3),(V),(Aiii) & AATCC TM70
Spray rating	See Table III	AATCC TM22
Resistance to organic liquid:	3.7.3	
Initial		AATCC TM118
After 5 launderings		AATCC TM135, (3,)(V),(Aiii) & AATCC TM118
Toxicity	3.8	See 4.7

1/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.
2/ Rated using the AATCC EP1, and AATCC EP2.
3/ Rated using the AATCC EP1.
4/ Rated using the AATCC EP8.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the finished cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.5.1). The finished cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

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4.6.2 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.
The rating shall be recorded. The results shall be recorded as “pass” or “fail.

4.7 Toxicity test. When the toxicity requirement (see 3.8) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (See 6.4 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Department or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the manufacturing of sleeping bags.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2.1).
c. When first article is required (see 3.1).
d. Color required (see 3.5).
e. Width of cloth required (see 3.10).
f. Length required if other than specified (see 3.11).
g. Inspection conditions (see 4.2).
h. Conformance inspection acceptance quality limits (AQLs) (see 4.3.2).
i. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)
Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250)

6.7 Subject term (key word) listing.

Fabric
Finish
N-tetradecane
Protection

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

(Project 8305-2023-001)

Review activities:
Army - MD
Navy – MC

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